

PROJECT PROPOSAL

StudyMate

A Personal Study Tracker & Analytics Application

Project Name	StudyMate — Study Tracker & Analytics
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Institution	Metropolitan University, Sylhet, Bangladesh
Technology	Python 3, Tkinter, Matplotlib
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01. INTRODUCTION

StudyMate is a lightweight desktop application that helps students track their daily study sessions, visualize their progress through charts, and review their study history — all from a clean graphical user interface. The application is built entirely using Python's standard library and one open-source charting package, making it accessible, portable, and beginner-friendly.

This project was designed as a practical demonstration of core Python programming concepts including functions, loops, lists, dictionaries, file handling, and conditional logic — all working together inside a real, usable piece of software.

02. PROBLEM STATEMENT

Many students study every day but have no way of knowing how much time they are actually spending on each subject. Without any record-keeping, it is difficult to identify weak areas, maintain consistency, or prepare for exams strategically. Existing solutions are either too complex, require internet access, or are not suited for simple academic use.

StudyMate solves this by providing:

- A simple way to log study sessions by subject and duration.
- A visual bar chart to compare time spent across subjects.
- A quick statistics summary showing total hours studied.
- A session log viewer to review all past records in one place.

03. PROJECT OBJECTIVES

- Build a functional desktop GUI application using Python and Tkinter.
- Allow users to log study sessions with subject name and minutes.
- Store session data in a plain text file for persistence across runs.
- Display a bar chart of total study time per subject using Matplotlib.
- Show summary statistics: total sessions, total hours, and minutes.
- Display all saved records inside the app without opening external files.
- Demonstrate practical use of Python fundamentals in a real project.

04. FEATURES & FUNCTIONALITY

Feature	Description
Save Session	User enters a subject name and number of minutes, which is appended to a log file and stored in memory.
Show Graph	Reads all sessions and draws a bar chart grouped by subject, showing total minutes studied per subject.
Statistics	Calculates and displays total number of sessions, total hours, and remaining minutes using integer division and modulus.
View Log	Opens a popup window listing all recorded sessions in reverse order so the most recent entry appears first.
Clear Records	Prompts the user for confirmation before wiping the log file, preventing accidental data loss.

05. TARGET USERS & WHY IT MATTERS

Primary Users

- School and university students who want to monitor their study habits.
- Students preparing for exams who need to balance time across subjects.
- Self-learners tracking progress in multiple topics simultaneously.

Why It Matters

Research consistently shows that self-monitoring improves academic performance. When students can see their data visually — which subject they are neglecting, how many hours they have put in this week — they are more likely to adjust their habits. StudyMate makes this kind of self-awareness effortless and available to anyone with a computer, even without an internet connection.

06. TECHNOLOGIES & TOOLS

Category	Tool / Library
Programming Language	Python 3 — core logic, data processing, file I/O
GUI Framework	Tkinter — built-in Python library for desktop GUI
Data Visualization	Matplotlib — bar charts and graph rendering
Data Storage	Plain text file (study_log.txt) — no database required
IDE / Editor	VS Code / PyCharm / IDLE — any Python-compatible editor
Version Control	Git — for tracking changes and submission

07. PYTHON CONCEPTS DEMONSTRATED

Every feature in StudyMate is built using basic Python fundamentals. The table below maps each language concept to its specific use in the project.

Python Concept	Where It Is Used in StudyMate
Functions	save_session(), show_graph(), show_stats(), view_log(), clear_log() — each feature is a separate function
for Loop	Iterating over sessions list to calculate totals; looping over file lines; creating buttons in a loop
if / else	Input validation — checking empty fields, non-numeric input, and empty session list before each action
List	sessions = [] stores all session tuples in memory during the program run
Tuple	Each session is stored as a tuple: (date, subject, minutes) inside the list
Dictionary	data = {} in show_graph() maps each subject name to its total minutes
File Handling	open(), write(), close() used in save_session() to persist data to study_log.txt
int() / str()	Converting user input string to integer with int(duration) before calculations

// and % operators	Integer division and modulus to convert total minutes into hours and remaining minutes
String methods	.strip() removes extra spaces from user input; .isdigit() validates numeric input

08. REQUIREMENTS TO BUILD THE PROJECT

Software Requirements

- Python 3.x installed on the machine (python.org).
- Tkinter — included by default with Python (no extra install needed).
- Matplotlib — install once via: `pip install matplotlib`

Hardware Requirements

- Any computer running Windows, macOS, or Linux.
- No internet connection required to run the application.
- No special hardware — runs on any standard student laptop or desktop.

Files Needed

- `studymate.py` — the single source file containing all code.
- `study_log.txt` — auto-created by the program when first session is saved.

09. CONCLUSION

StudyMate is a simple, practical, and self-contained Python application that solves a real problem for students. It is built entirely from core Python fundamentals — no advanced frameworks, no databases, no complex architecture. Every line of code can be explained and understood by a student who has completed an introductory Python course.

The project demonstrates that even fundamental programming concepts — functions, loops, lists, file handling, and conditionals — are powerful enough to build software that people can actually use every day.
